D. Make a Power of Two

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: greedy, math, strings | https://codeforces.com/problemset/problem/1560/D

You are given an integer n. In 1 move, you can do one of the following actions:

- erase any digit of the number (it's acceptable that the number before the operation has exactly one digit and after the operation, it is "empty");
- add one digit **to the right**.

The actions may be performed in any order any number of times.

Note that if, after deleting some digit from a number, it will contain leading zeroes, they will **not** be deleted. E.g. if you delete from the number 301 the digit 3, the result is the number 01 (not 1).

You need to perform the **minimum** number of actions to make the number any power of 2 (i.e. there's an integer k ($k \ge 0$) such that the resulting number is equal to 2^k). **The resulting number must not have leading zeroes**.

E.g. consider n=1052. The answer is equal to 2. First, let's add to the right one digit 4 (the result will be 10524). Then let's erase the digit 5, so the result will be 1024 which is a power of 2.

E.g. consider n=8888. The answer is equal to 3. Let's erase any of the digits 8 three times. The result will be 8 which is a power of 2.

Input

The first line contains one integer t ($1 \le t \le 10^4$) — the number of test cases. Then t test cases follow.

Each test case consists of one line containing one integer n ($1 \le n \le 10^9$).

Output

For each test case, output in a separate line one integer m — the minimum number of moves to transform the number into any power of 2.

Example

Input
12 1052 8888 6 75 128 1 301 12048 1504 6656 1000000000 687194767

Output

```
2
3
1
3
0
0
2
1
3
4
9
2
```

Note

The answer for the first test case was considered above.

The answer for the second test case was considered above.

In the third test case, it's enough to add to the right the digit 4 — the number 6 will turn into 64.

In the fourth test case, let's add to the right the digit 8 and then erase 7 and 5 — the taken number will turn into 8.

The numbers of the fifth and the sixth test cases are already powers of two so there's no need to make any move.

In the seventh test case, you can delete first of all the digit 3 (the result is 01) and then the digit 0 (the result is 1).